MOK ZHI ZHUAN

SOFTWARE DEVELOPER | PYTHON • DJANGO • FASTAPI • REACT • AI INTEGRATION

CONTACT

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Singapore
LinkedIn
Github
Portfolio

EDUCATION HISTORY

UOW Bachelor Games and Mobile Design

Computer Science

SIM(Singapore Institute of Management)

July 2018 - Dec 2021

461 Clementi Rd, Singapore 599491

Advanced Diploma of Information Technology

Information Technology(Computer Science , Security , System Analysis , etc)

MDIS(Management Development Institute of Singapore)

29/5/2017 - 29/12/2017

501 Stirling Rd, Singapore 148951

SKILLS

Programming Language

Python · Structured Text · JavaScript · Java · C++ · HTML/CSS

Backend & APIs

FastAPI · Django · REST API · MQTT

ABOUT

Software developer with hands-on experience building backend services, REST APIs, IoT monitoring platforms, and real-time computer vision applications. Experienced in Python, Django, FastAPI, React, PostgreSQL, and Docker, with practical exposure to YOLO-based object detection, OCR integration, and AI-assisted development workflows.

WORK EXPERIENCE

Software Application Developer

Winsys Technology PTE LTD

Feb 2024 - current / 18 Boon Lay Way, #03-120, TradeHub 21, 609966

- Developed automation and monitoring systems using CODESYS, Python, React, FastAPI/Django, and Tkinter.
- Implemented PLC communication and developed REST APIs for alarm handling and historical data storage, enhancing system reliability and data accessibility.
- Developed Python-based GUI and computer vision applications for a multi-camera system, enhancing real-time surveillance capabilities.
- Oversaw bug tracking for 20+ projects and coordinated testing and documentation processes using Jira, Notion, and Git.

Servicenow Consultant

NCS PTE LTD

August 2022 - December 2023 / 1 Depot Rd, Singapore 109679(DSTA)

- Enhanced service request resolution by 90% with an Automated Test Framework, incorporating detailed troubleshooting steps, and streamlined data configuration for onboarding teams.
- Collaborated with Senior Developers to assess Incident Reports, Change Requests, and Service Requests, resulting in a 50% reduction in issues identified in Javascript programming code updates.
- Completed training courses in CSA, ITIL, Cyber Security, and HackerRank, and trained over 10 system team members on system workflows, resulting in a 60% improvement in their ability to execute key processes.
- Collaborated in Agile development meetings to assess and improve developer activities, achieving a 50% reduction in task SLA and facilitating successful upgrades to ServiceNow versions Utah and Vancouver.

Peoplesoft Consultant

NCS Pte Ltd

Frontend
React.js · Tkinter · CODESYS HMI/WebVisu

Industrial Automation
CODESYS · PLC · Modbus RTU

Database
SQLite · SQL · Git/GitHub · Jira · Notion

Database / Tools
SQLite, SQL, Git, Jira, Notion

HOBBIES

Swimming
Programming
Jogging

MY KEY RESPONSIBILITIES

Software Development
Build and maintain automation, web, and Python applications.

System Integration
Integrate PLCs, APIs, databases, MQTT, and devices.

Testing & Troubleshooting
Test, troubleshoot, and improve system performance.

Project Collaboration
Handle requirements, documentation, Jira, and Git.

Jan 2022 - July 2022 /
5 Ang Mo Kio Street 62, NCS Hub Singapore 569141

- Completed PeopleSoft training, which improved payroll processing efficiency by 20% through optimized data entry and quicker access to employee records.
- Modified Peoplesoft programs to implement new features, resulting in a 70% reduction in processing time for client report generation.
- Gained hands-on experience with the software development life cycle while modifying 10+ Peoplesoft programs, enhancing features by 70%.
- Participated in weekly team meetings to review and enhance developer activities, proposing two key strategies that increased project efficiency by 15% during the AGD Development process.

TECHNICAL PROJECTS

Industrial Analyzer Monitoring System

CODESYS, MQTT, Modbus RTU, SQLite, HMI

May 2026 - Sep 2026

- Developed a 16-channel industrial analyzer monitoring system using **CODESYS** for real-time data processing and visualization.
- Integrated **Modbus RTU, MQTT, alarm handling, historical records, calibration, and local database storage**.
- Developed HMI pages for system status, alarms, communication logs, calibration, and configuration settings.

AI People Counting & Multi-Camera System

Python, YOLO, OpenCV, React, FastAPI/Django

Sep 2026 - current

- Developed a computer-vision people counting system for **entry, exit, and occupancy monitoring**.
- Integrated camera processing with **REST APIs, database storage, and a React-based dashboard**.
- Designed the application to support **multiple cameras** and separate camera configurations.

GreyForm PBU / BIM Robotic Marking System

Python, IFC/BIM, tktinker , ROS, Fastapi

April 2021 - Aug 2021

- Developed a Python application to process **IFC/BIM geometry** for prefabricated bathroom unit manufacturing.
- Built a desktop interface for visualizing and processing wall geometry and marking coordinates.